Intake Test

Does the intake pull balls in, does the water wheel lift them, and do the two switches spot a NECTAR? (issue #4, decisions 024, 025, 028)

Names		Date	
Battery voltage at start		Battery at end	

OpMode: Intake Test (Tests group on the Driver Station). **Needs in the config:** `intakeServo` (Continuous Rotation Servo), `waterWheelMotor` (goBILDA 5203 motor), `nectarDetectorLeftSwitch`, `nectarDetectorRightSwitch` (Digital Device)

Hidden by default (052): delete `@Disabled` in its file, build, deploy. **Safety:** space around the robot, hands clear, one person on STOP. Nothing moves in INIT.

Controls (gamepad 1)

Button	Does
A / B	spin ON-OFF (toggle) / STOP. Starts stopped
DPAD up / down	speed +/- 0.05 of whichever is running (BOTH = both). Intake and water wheel each have their own speed (the water wheel is a motor since 11/11). Not saved: copy the numbers into Intake.java
DPAD left / right	direction (toggle): IN + water wheel UP, or OUT + water wheel DOWN
X	which one runs: BOTH, INTAKE only, WATER WHEEL only. Use "only" to check one at a time
(no button)	the switches are read all the time, even in INIT. Watch the telemetry lines "NECTAR detected", "Nectar left switch", "Nectar right switch"

Steps, in order

#	Do this	Expected	Actual (what happened)	If wrong
1	X to INTAKE only, press A (direction IN)	balls go IN		`INTAKE_REVERSED = true`
2	X to WATER WHEEL only, press A (direction UP)	balls go UP toward the launcher		`WATER_WHEEL_REVERSED = true`
3	Tune each speed with X on its "only" setting and dpad up / down, then X to BOTH, A on, 3 balls	no jam, no ball flying out		copy the two speeds from the screen into `INTAKE_SPEED` / `WATER_WHEEL_SPEED` (0 to 1)
4	Nothing touching the levers	both "not pressed", "NECTAR detected: no"		says PRESSED: switch `NECTAR_SWITCH_WIRING` to the other value

#	Do this	Expected	Actual (what happened)	If wrong
5	Press the LEFT lever by hand	"Nectar left switch: PRESSED", "NECTAR detected: YES"		never changes: try the other channel of that digital port (n / n+1)
6	Press the RIGHT lever by hand	same, for right		same
7	Run a NECTAR (big ball, ~3.6 in) through	YES while it passes		levers too far apart: move them in
8	Run a POLLEN (small ball, ~2.8 in) through	stays "no"		levers too close: move them out

Step 7 and 8, repeat 5 times each. NECTAR detected: ___ / 5. POLLEN wrongly detected: ___ / 5.

Numbers to put into the code: `subsystems/Intake.java`

Constant	Was	New value	Checked by
`INTAKE_REVERSED`	true		
`WATER_WHEEL_REVERSED`	true		
`INTAKE_SPEED`	1.0		
`WATER_WHEEL_SPEED`	1.0		
`NECTAR_SWITCH_WIRING`	NORMALLY_CLOSED		

Wiring, for `hardware-config.md` (Intake table)

Device	Hub (CH / EH)	Port
`intakeServo`		servo
`waterWheelMotor`		motor
`nectarDetectorLeftSwitch`		digital
`nectarDetectorRightSwitch`		digital

After the test: a coder puts the new numbers into the constants and removes the matching TODO comments; the results go on GitHub issue #4. The filled-in sheet goes with the portfolio entry `docs/portfolio/2026-11-04-intake-and-nectar-detection.md`. Photos of anything surprising!